

Did you code yesterday?

Did you try coding the VSCode landing page yesterday?

Shoutouts -

1. <https://x.com/CtrlAltElite111/status/1820076637477564416>
2. <https://x.com/BuggyNaman/status/1820060663319769462>
3. https://x.com/import_the_code/status/1820082443506114582
4. https://x.com/varshaa_dev/status/1820085647190712649
5. https://x.com/_ _Raiders/status/1820089916287828123
6. https://x.com/bharat__2044/status/1820090993045020979

Bounty - \$25 to each of you!

Javascript – The basics

Web development

Web development involves writing a lot of HTML, CSS and JS code.

Historically (and even today to some extent), browsers could only understand HTML, CSS and JS

Any website that you see, is a bunch of HTML, CSS and JS files along with some assets (images, videos etc)

Facts/Callouts

1. React, NextJS are **frameworks** . They compile down to HTML, CSS, JS in the end. That is what your browser understands.
2. When you run your C++ code on **leetcode** , it does not run on your browser/machine. It runs somewhere else. Your browser can't (almost) compile and run C++ code.
3. If someone asks — What all languages can your browser interpret, the answer is HTML, CSS, JS and WebAssembly. It can, technically, run C++/Rust code that is compiled down to Wasm

Before we proceed, do one of the following –

1. Create an account on replit
2. Install Node.js locally

Properties of JS

Every language comes with its unique set of features.

Javascript has the following -

1. Interpreted

JavaScript is an interpreted language, meaning it's executed line-by-line at runtime by the JavaScript engine in the browser or server environment, rather than being compiled into machine code beforehand.

Upsides -

1. There is one less step to do before running your code

Downsides -

1. Performance Overhead:
2. More prone to runtime errors

2. Dynamically Typed

Variables in JavaScript are not bound to a specific data type. Types are determined at runtime and can change as the program executes

C++ Code (won't compile)

```
#include <iostream>
```



```
int main() {  
  int a = 1;  
  a = "hello";  
  a = true;  
}
```

JS Code (will compile)

```
var a = 1;  
a = "harkirat";  
a = true;  
  
console.log(a)
```



3. Single threaded

JavaScript executes code in a single-threaded environment, meaning it processes one task at a time. We will dive deeper into this next week.

4. Garbage collected

JavaScript automatically manages memory allocation and deallocation through garbage collection, which helps prevent memory leaks by automatically reclaiming memory used by objects no longer in use.

Conclusion

Is JS a good language?

Yes and no. It is beginner friendly, but has a lot of performance overhead. **Bun** is trying to solve for a lot of this, but there's a long way to go before JS can compete with languages like C++/Rust

Syntax of Javascript

1. Variables

Variables are used to store data. In JavaScript, you declare variables using `var`, `let`, or `const`.

```
let name = "John"; // Variable that can be reassigned
const age = 30;    // Constant variable that cannot be reassigned
var isStudent = true; // Older way to declare variables, function-scoped
```



► Assignment

2. Data types

```
let number = 42; // Number
let string = "Hello World"; // String
let isActive = false; // Boolean
let numbers = [1, 2, 3]; // Array
```



3. Operators

```
let sum = 10 + 5; // Arithmetic operator
let isEqual = (10 === 10); // Comparison operator
let isTrue = (true && false); // Logical operator
```



4. Functions

```
// Function declaration
function greet(name) {
  return "Hello, " + name;
}

// Function call
let message = greet("John"); // "Hello, John"
```



► Assignment #1

► Assignment #2

5. If/Else

```
if (age >= 18) {  
  console.log("You are an adult.");  
} else {  
  console.log("You are a minor.");  
}
```



► Assignment

6. Loops

```
// For loop  
for (let i = 0; i < 5; i++) {  
  console.log(i); // Outputs 0 to 4  
}  
  
// While loop  
let j = 0;  
while (j < 5) {  
  console.log(j); // Outputs 0 to 4  
  j++;  
}
```



► Assignment

Complex types

Objects

An object in JavaScript is a collection of **key-value pairs**, where each **key** is a string and each **value** can be any valid JavaScript data type, including another object.

```
let user = {  
  name: "Harkirat",  
  age: 19  
}  
  
console.log("Harkirats age is " + user.age);
```



- ▶ Assignment #1
- ▶ Assignment #2
- ▶ Assignment #3

Arrays

Arrays let you group data together

```
const users = ["harkirat", "raman", "diljeet"];  
const totalUsers = users.length;  
const firstUser = users[0];
```



► Assignment

Array of Objects

We can have more complex objects, for example an array of objects

```
const users = [{
  name: "Harkirat",
  age: 21
}, {
  name: "raman",
  age: 22
}]

const user1 = users[0]
const user1Age = users[0].age
```



► Assignment

Object of Objects

We can have an even more complex object (object of objects)

```
const user1 = {
  name: "harkirat",
  age: 19,
  address: {
    city: "Delhi",
    country: "India",
    address: "1122 DLF"
  }
}

const city = user1.address.city;
```



► Assignment

